

Clinical Variant Interpretation Report

Report metadata

Report schema: 1.0 Evidence schema: 1.0 Interpretation prompt: 1.1 LLM model: gpt-5.4-mini Assembly: GRCh38

Case Summary

- Provided HPO terms: HP:0034270
- Matched HPO terms: Not available in the supplied evidence.

Variant Summary

- Assembly: GRCh38
- Variant: MT:14560 G>C
- Population frequency: Not available in the supplied evidence.

Gene and Consequence

- Gene: MT-ND6
- Gene ID: ENSG00000198695
- Transcript: ENST00000361681
- Consequence: synonymous_variant
- Impact: LOW
- Protein change: ENSP00000354665.2:p.Val38=

Clinical Evidence

ClinVar

- Accession: Not available in the supplied evidence.
- Clinical significance: Not available in the supplied evidence.
- Review status: Not available in the supplied evidence.
- Conditions: Not available in the supplied evidence.

ClinGen

- Disease: Leigh syndrome; Disease ID: MONDO:0009723; Classification: Definitive; Mode of inheritance: Mitochondrial
- Disease: mitochondrial disease; Disease ID: MONDO:0044970; Classification: Definitive; Mode of inheritance: Mitochondrial

Phenotype Correlation

- Phenotype score: 0
- Provided HPO terms: HP:0034270
- Matched HPO terms: Not available in the supplied evidence.

Interpretation

- The evidence object provides gene-disease validity evidence for MT-ND6, but it does not provide a variant-specific clinical classification for this MT-ND6 synonymous variant.
- ClinVar and MyVariant.info returned no exact result for this variant.
- Based on the supplied evidence alone, a definitive variant interpretation is unsupported.

Limitations

- Variant-specific pathogenicity evidence is not provided.
- No ClinVar accession, review status, or significance is available.
- No population frequency is available.
- The meaning of HP:0034270 is not expanded in the supplied evidence.
- Warnings note that MyVariant.info and NCBI ClinVar returned no exact result for this variant.

Evidence source availability

- vep: success
- myvariant: not_found
- clinvar: not_found
- clingen: success

Report warnings

- MyVariant.info returned no exact result for this variant.
- NCBI ClinVar returned no exact result for this variant.

References

- Ensembl VEP - https://rest.ensembl.org/vep/homo_sapiens/region
- ClinGen Gene-Disease Validity via UCSC GenCC - https://search.clinicalgenome.org/kb/gene-validity/CGGV:assertion_836f975f-27f5-47c9-a58e-35089b6b39e6-2021-05-17T04:00:00.000Z
- ClinGen Gene-Disease Validity via UCSC GenCC - https://search.clinicalgenome.org/kb/gene-validity/CGGV:assertion_7f581c96-e3cc-49bb-aa2b-98650efa4157-2023-05-03T16:00:00.000Z
- ClinGen - MONDO:0009723 - https://search.clinicalgenome.org/kb/gene-validity/CGGV:assertion_836f975f-27f5-47c9-a58e-35089b6b39e6-2021-05-17T04:00:00.000Z
- PubMed - PMID:8622678
- PubMed - PMID:10894222
- PubMed - PMID:14595656
- PubMed - PMID:14684687
- PubMed - PMID:15576045
- PubMed - PMID:17535832
- PubMed - PMID:19062322
- PubMed - PMID:19103152
- PubMed - PMID:20019223
- PubMed - PMID:21196529
- PubMed - PMID:21749722
- PubMed - PMID:23129651

- PubMed - PMID:23813926
- PubMed - PMID:25356405
- PubMed - PMID:27509854
- PubMed - PMID:33536343
- ClinGen - MONDO:0044970 - https://search.clinicalgenome.org/kb/gene-validity/CGGV:assertion_7f581c96-e3cc-49bb-aa2b-98650efa4157-2023-05-03T16:00:00.000Z
- PubMed - PMID:2018041
- PubMed - PMID:11133798
- PubMed - PMID:21364701
- PubMed - PMID:30741831

Medical Disclaimer

AI-generated decision-support summary based only on the supplied Evidence Object. It is not a diagnosis or treatment recommendation and requires review by a qualified healthcare professional.